

The Missing Powerhouse

I PUC · Ch 8 · Cell: The Unit of Life

Starring Maya 👧 & Professor Nucleus 🔬

A BAIO storybook by Narendra Sir · Biology Lecturer



Maya’s Drooping Sunflower

Maya’s sunflower had water, sunlight and good soil — yet its leaves were pale and weak. Suddenly a tiny scientist in a purple coat landed on a leaf. "I am Professor Nucleus! Every living thing is made of cells — that is the **cell theory** of **Schleiden** and **Schwann**. And as **Virchow** said, all cells come from pre-existing cells. Let’s go inside and find the problem!"

💡 **REMEMBER**

Cell theory: all living organisms are composed of cells and products of cells; all cells arise from pre-existing cells.

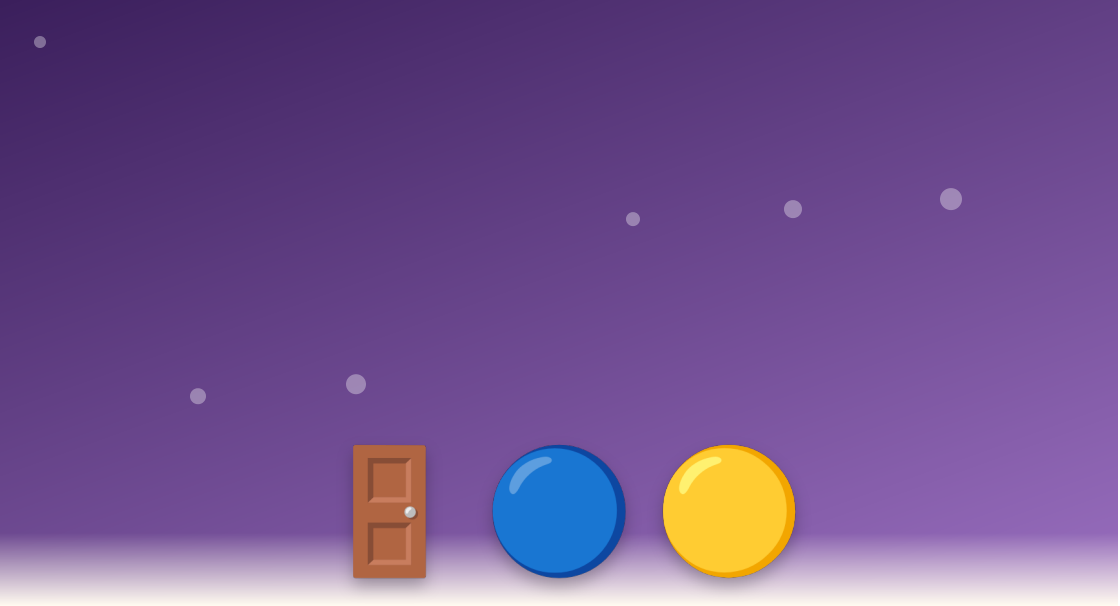


Through the Wall

They shrank and reached a strong green wall. "This is the **cell wall**," said the professor. "Made of **cellulose**, hemicellulose, pectins and proteins." Between two cells lay a sticky layer of calcium pectate — the **middle lamella**. Tiny tunnels called **plasmodesmata** crossed the walls, joining the cytoplasm of neighbouring cells. Maya's first task: name it! The cell glowed a little brighter.

💡 **REMEMBER**

The primary wall of a young cell can grow; the secondary wall forms later on its inner side.



The Selective Gatekeeper

Next came a soft, moving gate — the **plasma membrane**. "It is a **lipid bilayer** with proteins floating in it," explained the professor. "Singer and Nicolson called it the **fluid mosaic model** (1972)." Water slipped through by **osmosis**, while some ions were pumped in using energy — **active transport**. "It is **selectively permeable**," Maya said proudly. Another light switched on!

💡 **REMEMBER**

Passive transport needs no energy (diffusion, osmosis); active transport uses ATP, e.g. the Na^+/K^+ pump.

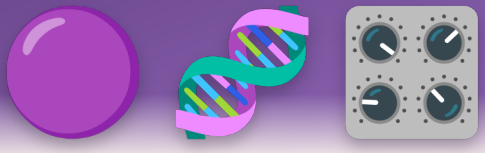


The Jelly Sea

Inside, they floated in a jelly-like **cytoplasm**, the site of many reactions. Granules called **ribosomes** were busy making proteins. A maze of tubes — the **endoplasmic reticulum** — ran through the cell: **rough ER** studded with ribosomes, and **smooth ER** that makes lipids. Stacks of flat sacs, the **Golgi apparatus**, packed products into little bubbles.

💡 **REMEMBER**

Eukaryotic ribosomes are 80S (60S + 40S). Golgi is the site of glycoprotein and glycolipid formation.



The Control Room

A giant purple sphere loomed ahead. "My home — the **nucleus**, first seen by **Robert Brown**!" It had a double **nuclear envelope** with **nuclear pores**. Inside, thread-like **chromatin** carried DNA, and a dense **nucleolus** made ribosomal RNA. "The nucleus controls every activity of the cell," said Maya. "Correct!" The control room lights blinked on.

💡 **REMEMBER**

Mammalian red blood cells and sieve tube cells of plants have no nucleus.

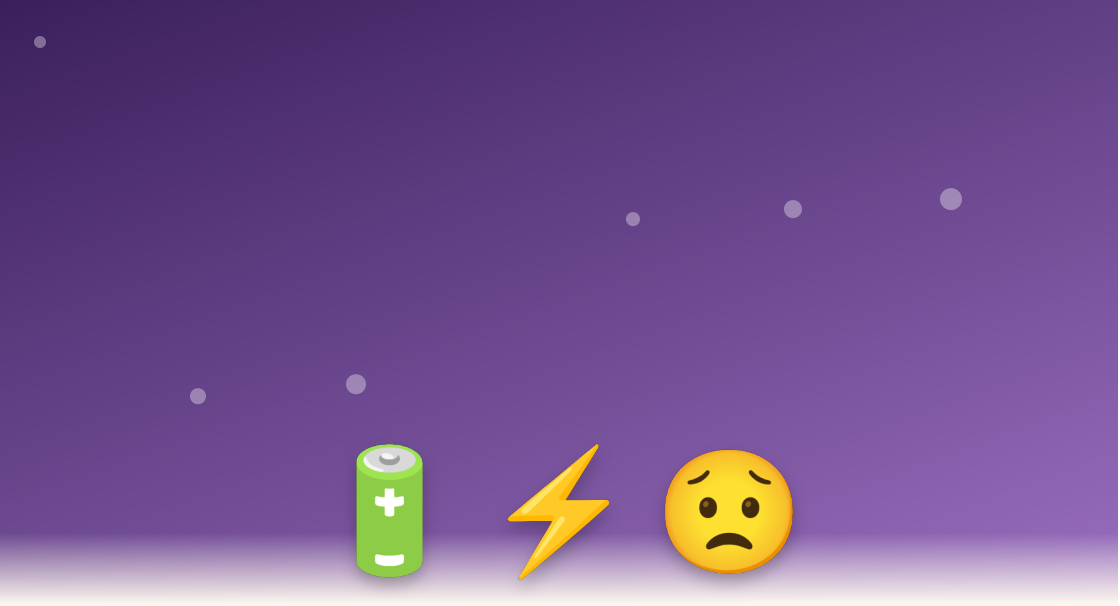


Kitchens and a Water Tank

Green discs drifted past — **chloroplasts**. Stacks of **thylakoids** called **grana** trapped sunlight, and the fluid **stroma** held enzymes for making sugar. In the centre lay a huge water tank, the **vacuole**, wrapped in a membrane called the **tonoplast**. "In plant cells it can fill up to 90 percent of the volume," the professor said.

💡 **REMEMBER**

Plastids: chloroplasts (green), chromoplasts (yellow-orange-red carotenoids) and leucoplasts (colourless stores).



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The Silent Powerhouse

Finally they found the problem: sausage-shaped **mitochondria** lying silent. "These are the powerhouses — sites of **aerobic respiration** that make **ATP**," said the professor. Maya studied one: an outer membrane, an inner membrane folded into **cristae**, and a **matrix** with its own circular DNA and 70S ribosomes. As she named each part, the cristae began to hum.

💡 REMEMBER

Mitochondria are semi-autonomous: they have their own DNA and ribosomes and divide by fission.



The Sunflower Stands Tall

With every organelle named, the mitochondria roared back to life and ATP flowed through the cell like electricity. Back outside, Maya watched her sunflower lift its head toward the sun, its leaves deep green again. "Remember," said Professor Nucleus, "a cell is not just a bag — it is a city where every part has a job." And with a wink, he flew away.

💡 **REMEMBER**

Prokaryotic cells lack a nuclear membrane and membrane-bound organelles; eukaryotic cells have both.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

cell theory p.1 · Schleiden p.1 · Schwann p.1 · Virchow p.1

cell wall p.2 · cellulose p.2 · middle lamella p.2 · plasmodesmata p.2

plasma membrane p.3 · lipid bilayer p.3 · fluid mosaic model p.3

osmosis p.3 · active transport p.3 · selectively permeable p.3

cytoplasm p.4 · ribosomes p.4 · endoplasmic reticulum p.4

rough ER p.4 · smooth ER p.4 · Golgi apparatus p.4 · nucleus p.5

Robert Brown p.5 · nuclear envelope p.5 · nuclear pores p.5

chromatin p.5 · nucleolus p.5 · chloroplasts p.6 · thylakoids p.6

grana p.6 · stroma p.6 · vacuole p.6 · tonoplast p.6

mitochondria p.7 · aerobic respiration p.7 · ATP p.7 · cristae p.7

matrix p.7



QUICK QUIZ

Do you remember?

1. Which organelle is called the powerhouse of the cell?

- Ribosome
- Mitochondrion
- Golgi apparatus
- Lysosome

2. Which organelle controls the activities of the cell?

- Vacuole
- Nucleus
- Chloroplast
- Cell wall

3. Which structure controls what enters and leaves the cell?

- Cell wall
- Plasma membrane
- Nucleolus
- Cytoplasm

4. The fluid mosaic model was proposed by:

- Watson and Crick
- Singer and Nicolson
- Schleiden and Schwann
- Robert Brown

5. The membrane around the plant vacuole is the:

- Tonoplast
- Middle lamella
- Nuclear envelope
- Cristae

Answers: 1-Mitochondrion · 2-Nucleus · 3-Plasma membrane · 4-Singer and Nicolson · 5-Tonoplast